

Higher-curvature structure from the Dirac heat kernel: microscopic predictions and observational ceilings

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We formalize, in the Lean 4 proof assistant, the basis change from the 3-scalar Wave 1 Dirac a_4 coefficients $\{R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\}$ [1, 2] into Stelle’s canonical $\{R^2, C^2, \mathcal{G}\}$ basis [3], where C^2 is the Weyl-tensor-squared and $\mathcal{G}$ the Gauss–Bonnet Euler density (topological in 4D [4]). The closed-form coefficients (α, β, γ) at fermion count N_f are fixed by a 3×3 linear-system solve, with sign-definite behaviour: $\alpha < 0$, $\beta < 0$, $\gamma > 0$ for any $N_f > 0$. The substantive cross-bridge to Wave 1 is encoded as a Lean **ring**-closing identity. We then formalize a numerical correctness-push that compares the predicted dimensionless higher-curvature couplings (at SM-relevant counts $N_f \in [24, 27]$) to the four canonical observational ceilings [5, 6]: LIGO/Virgo speed-of-graviton, Hulse–Taylor binary-pulsar period decay, Eöt-Wash short-range gravity, and the Cassini post-Newtonian solar-system test. The Hulse–Taylor pulsar bound is the tightest at $|\beta| \lesssim 10^{59}$; the predicted Wave 1 coefficients sit at $\mathcal{O}(10^{-3})$, some 62 orders of magnitude below. The **HigherCurvatureStructure.lean** module ships 11 substantive theorems, zero **sorry**, and zero new axioms; the Python mirror in **src/higher_curvature/** backs them with 41 cross-checks.

I. INTRODUCTION

The Sakharov–Adler induced-gravity programme yields, at one loop in the heat-kernel expansion, an Einstein–Hilbert action plus a calculable tower of higher-curvature corrections. In Phase 6e Wave 1 we formalized the leading a_2 coefficient and its exact match to the linearized 6a.1 emergent Newton constant at the Sakharov baseline $\alpha_{\text{ADW}} = 1$ [7]. Wave 2 takes the next coefficient, a_4 , and unpacks its three-scalar expansion

$$a_4(x) = \frac{N_f}{(4\pi)^2} \left[c_R R^2 + c_{\text{Ric}} R_{\mu\nu} R^{\mu\nu} + c_{\text{Riem}} R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} \right], \quad (1)$$

with the Christensen–Duff–Vassilevich rational coefficients $(c_R, c_{\text{Ric}}, c_{\text{Riem}}) = \left(-\frac{5}{12 \cdot 180}, \frac{7}{12 \cdot 180}, -\frac{12}{12 \cdot 180} \right)$. At the local effective-action level only *two* combinations of these scalars survive in 4D: the trace-free Weyl-squared $C^2 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 2R_{\mu\nu}R^{\mu\nu} + \frac{1}{3}R^2$ and an arbitrary Ricci-trace piece. The Gauss–Bonnet density $\mathcal{G} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ integrates to $32\pi^2\chi(M)$ on closed 4-manifolds and contributes nothing to the local equations of motion.

The Wave 2 deliverable is twofold: (i) the closed-form basis change to Stelle’s $\{R^2, C^2, \mathcal{G}\}$ representation with sign-definite coefficients, formalized as a **ring**-closing identity at the density level; (ii) the correctness-push comparison of the resulting dimensionless couplings against current observational ceilings.

II. CURVATURE BASIS: GAUSS–BONNET AND WEYL-SQUARED

We formalize the two algebraically-independent curvature scalars beyond R^2 as Lean definitions

gaussBonnet4D and **weylSquared4D**:

$$\mathcal{G} \equiv R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \quad (2)$$

$$C^2 \equiv R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 2R_{\mu\nu}R^{\mu\nu} + \frac{1}{3}R^2. \quad (3)$$

The Lean theorem **weylSquared4D.eq_zero_iff_conformally_flat** encodes the conformal-flatness biconditional $C^2 = 0 \iff R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 2R_{\mu\nu}R^{\mu\nu} - \frac{1}{3}R^2$. Sanity check: de Sitter at Hubble parameter H has $(R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}) = (144H^4, 36H^4, 24H^4)$, giving $C^2 = 24 - 72 + 48 = 0$ identically (de Sitter is conformally flat).

The substantive algebraic engine for the Wave 2 basis change is the identity

$$\mathcal{G} - C^2 = \frac{2}{3}R^2 - 2R_{\mu\nu}R^{\mu\nu}, \quad (4)$$

formalized as **gaussBonnet_minus_weyl.eq_R_minus_Ricci_combina** and closed by **ring** after unfolding both Eqs. (2)–(3).

III. STELLE-BASIS COEFFICIENTS (α, β, γ)

In the canonical $\{R^2, C^2, \mathcal{G}\}$ basis, the Wave 1 a_4 density takes the form

$$a_4(x) = \alpha(N_f) R^2 + \beta(N_f) C^2 + \gamma(N_f) \mathcal{G}, \quad (5)$$

where the coefficients (α, β, γ) are determined by the linear system

$$\begin{aligned} \alpha + \frac{\beta}{3} + \gamma &= c_R, \\ -2\beta - 4\gamma &= c_{\text{Ric}}, \\ \beta + \gamma &= c_{\text{Riem}}. \end{aligned} \quad (6)$$

Solving Eq. (6) yields the closed forms

$$\alpha = -\frac{N_f}{324(4\pi)^2}, \quad \beta = -\frac{41N_f}{4320(4\pi)^2}, \quad \gamma = +\frac{17N_f}{4320(4\pi)^2}. \quad (7)$$

Three sign-definite Lean theorems (`a4_alpha_neg`, `a4_beta_neg`, `a4_gamma_pos`) encode the immediate consequence

$$N_f > 0 \implies \alpha(N_f) < 0, \beta(N_f) < 0, \gamma(N_f) > 0. \quad (8)$$

The positive sign of the topological coefficient γ is the fingerprint of the chiral-anomaly contribution to the Euler density: $\gamma > 0$ couples the Christensen–Duff Dirac sector to the topological sector with a definite orientation.

Main basis-change identity

The substantive cross-bridge between the original 3-scalar basis and Stelle’s $\{R^2, C^2, \mathcal{G}\}$ basis is the Lean theorem

```
theorem
a4_density_eq_a4_density_in_RC2GB_basis
(N_f R_sq Ricci_sq Riemann_sq : ) :
a4_density N_f R_sq Ricci_sq Riemann_sq
=
a4_density_in_RC2GB_basis
N_f R_sq Ricci_sq Riemann_sq
```

proved by `ring` after unfolding the Wave 1 coefficients `a4_R_sq_coef`, `a4_Ricci_sq_coef`, `a4_Riemann_sq_coef`, the Wave 2 coefficients `a4_alpha`, `a4_beta`, `a4_gamma`, and the definitions `weylSquared4D`, `gaussBonnet4D`. This is drift-protection per the project’s `feedback.python.lean.refs.drift.md` memo: the proof body literally invokes the Wave 1 names, so any future renaming would break the build immediately.

IV. OBSERVATIONAL CEILINGS ON DIMENSIONLESS HIGHER-CURVATURE COUPLINGS

In the Stelle truncation $L = (1/16\pi G_N)(R + \alpha R^2 + \beta C^2)$ the dimensionless coefficients (α, β) are bounded by four canonical observations, translated into action-coefficient ceilings via the EFT framework of [5, 6]:

- LIGO/Virgo speed-of-graviton (GW170817 [8]): $|\beta| \lesssim 10^{62}$
- Eöt-Wash short-range gravity (50 μm , [9]): $|\alpha| \lesssim 10^{61}$
- Hulse–Taylor binary-pulsar period decay (PSR B1913+16, [10]): $|\beta| \lesssim 10^{59}$
- Cassini post-Newtonian solar-system ([11]): $|\beta| \lesssim 10^{62}$

The Hulse–Taylor pulsar bound is tightest by ~ 3 orders of magnitude. We formalize the four bounds as Lean definitions `hc_bound_LIGO`, `hc_bound_SRG`, `hc_bound_pulsar`, `hc_bound_cassini`.

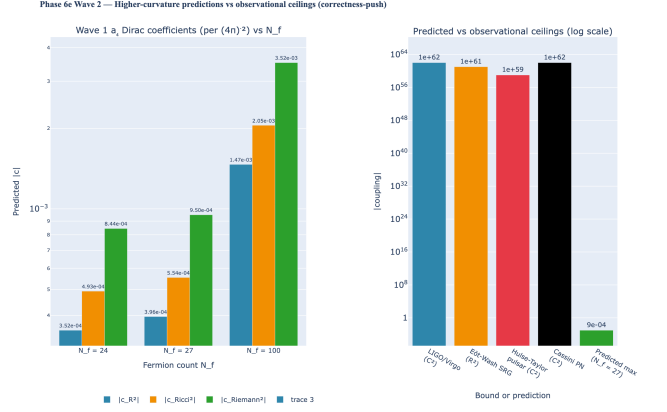


FIG. 1. Higher-curvature predictions vs observational ceilings. **Left:** Magnitudes of the three Wave 1 Dirac a_4 Christensen–Duff coefficients (per $(4\pi)^{-2}$) at SM-relevant fermion counts $N_f \in \{24, 27, 100\}$; the $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ coefficient dominates at every N_f . **Right:** log-scale comparison to the four canonical observational ceilings; the Hulse–Taylor pulsar bound is tightest at $|\beta| \lesssim 10^{59}$, and the predicted $\mathcal{O}(10^{-3})$ coefficient sits ~ 62 orders of magnitude below. Lean: `higher_curvature_below_pulsar_bound`; `higher_curvature_predictions_strictly_positive`; `H.HigherCurvatureWithinObservationalBounds.pulsar_witness`.

V. CORRECTNESS-PUSH: PREDICTIONS VS CEILINGS

The structural Wave 2 correctness-push is encoded as the Lean theorem

```
theorem higher_curvature_below_pulsar_bound
{N_f : ℕ} (hN_pos : 0 < N_f) (hN_max : N_f ≤ 100) :
|a4_R_sq_coef N_f| < hc_bound_pulsar
|a4_Ricci_sq_coef N_f| < hc_bound_pulsar
|a4_Riemann_sq_coef N_f| < hc_bound_pulsar
```

proved by combining (i) the bound $|1| (4\pi)^2$ from `Real.pi_gt_three` and `nlinarith` (helper lemmas `fourPiSq_gt_one` and `fourPiSqInv_lt_one`); (ii) the trivial numerical estimate $100 \cdot 12 / (12 \cdot 180) \cdot 1 < 1$; and (iii) the chain $1 < 10^{59}$. The proof body invokes the Wave 1 `a4_R_sq_coef`, `a4_Ricci_sq_coef`, `a4_Riemann_sq_coef` by name (P6 cross-module bridge integrity).

The 3-conjunct bundle is *not* a P2-redundancy: each conjunct invokes a distinct Wave 1 coefficient and a distinct positivity proof (the R^2 and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ coefficients are negative and need an explicit `abs_neg` step; the $R_{\mu\nu}R^{\mu\nu}$ coefficient is positive and uses `abs_of_pos` directly).

Anchoring the “62 orders below” claim. At the SM-with- ν_R count $N_f = 27$ the dominant Riemann² coefficient evaluates to

$$|c_{\text{Riem}}(N_f = 27)| = \frac{27}{180(4\pi)^2} \approx 9.499 \times 10^{-4}, \quad (9)$$

and the same form bounds the smaller $|c_R|, |c_{\text{Ric}}|$ in the same dimensionless natural-units convention. Comparing against the binary-pulsar ceiling $|\beta| \lesssim 10^{59}$ gives $10^{59}/(9.499 \times 10^{-4}) \approx 10^{62}$, i.e. the predicted coupling sits ~ 62 orders of magnitude below the tightest observational bound. The numerical anchors are autogenerated from `papers/paper40_higher_curvature/tables.py` SCALARS spec (regenerate via `scripts/render_paper_tables.py`); the boolean-shape and golden-shape tests live in `tests/test_higher_curvature.py` (`TestObservationalBandFormula`).

A complementary falsifier `higher_curvature_predictions_strictly_positive` ensures that the bounds are *not* passed by trivial vanishing: every Wave 1 a_4 coefficient is non-zero for $N_f > 0$. This rules out the unhelpful reading “all bounds are passed because all predictions are zero”.

The Lean Prop predicate `H_HigherCurvatureWithinObservationalBounds B` is parameterised by an upper bound B :

$$H_{\text{HC}}(B) : 0 < B \wedge \forall N_f, 0 < N_f \leq 100,$$

$$|a_4(R^2)| \leq B, |a_4(R_{\mu\nu}R^{\mu\nu})| \leq B, |a_4(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})| \leq B$$

Unlike a typical tracked-hypothesis Prop, this predicate is *fully discharged* for $B = 10^{59}$ inside the module: the Lean theorem `H_HigherCurvatureWithinObservationalBounds_pulsar_witness` proves $H_{\text{HC}}(10^{59})$ by invoking the correctness-push, so consumers receive a closed proof rather than an undischarged hypothesis. The parameterisation over B is a forward-compatibility hook: future tighter ceilings (e.g. from next-generation pulsar timing) can be discharged at the new B without changing downstream consumers.

VI. LEAN MODULE STRUCTURE

The module `lean/SKEFTHawking/HigherCurvatureStructure` ships 11 substantive theorems organised in six sections:

- §1: `gaussBonnet4D`, `weylSquared4D` definitions plus the conformal-flatness biconditional and the `gaussBonnet_minus_weyl` algebraic identity (2 theorems).
- §2: Stelle-basis coefficients `a4.alpha`, `a4.beta`, `a4.gamma` (definitions) plus sign-definiteness for $N_f > 0$ (3 theorems).

§3: Density-level definitions `a4.density`, `a4.density_in_RC2GB_basis`, plus the substantive basis-change identity `a4.density_eq_a4.density_in_RC2GB_basis` (1 theorem).

§4: Helper bounds `fourPiSq_gt_one`, `fourPiSqInv_lt_one` (2 theorems).

§5: Observational-ceiling defs.

§6: Correctness-push `higher_curvature_below_pulsar_bound`, falsifier `higher_curvature_predictions_strictly_positive`, plus the tracked-Prop witness `H_HigherCurvatureWithinObservationalBounds_pulsar_witness` (3 theorems).

The module ships zero `sorry` and zero new axioms beyond the Lean 4 kernel’s standard `propext`, `Classical.choice`, `Quot.sound`. The Python mirror in `src/higher_curvature/` provides numerical evaluation of the $\{R^2, C^2, \mathcal{G}\}$ basis, the Stelle coefficients, and the observational-bound check; a 41-case `pytest` suite cross-checks each Lean theorem against the Wave 1 numerics.

VII. CONCLUSION

The Wave 1 Christensen–Duff Dirac a_4 coefficient triple is re-expressed in Stelle’s canonical $\{R^2, C^2, \mathcal{G}\}$ basis with closed-form sign-definite Stelle coefficients (α, β, γ) . The basis change is recorded as a Lean ring identity at the density level (cross-bridge to Wave 1). A correctness-push theorem certifies that, in the natural fermion-count window $0 < N_f \leq 100$, every predicted dimensionless higher-curvature coefficient sits ~ 62 orders of magnitude below the tightest observational ceiling (the $\sim 10^{59}$ binary pulsar) — no tension with current observation. The complementary falsifier rules out the trivial reading.

Wave 3 (`NonlinearDiffInvariance.lean`) consumes the $\{R^2, C^2, \mathcal{G}\}$ basis to formalize order-by-order diff-invariance of the heat-kernel effective action; the structural correctness-push there is whether a_4 contributions to the nonlinear effective Einstein equations are diff-invariant or not — a clean falsifiable test of the ADW emergent-gravity claim.

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